Recoverability Frontiers for Internal Correspondence After Logic Synthesis

A proof-carrying artifact for AIG optimization experiments

AIG Optimization Experiments

August 1, 2026

Abstract

Logic synthesis preserves input-output behavior while deliberately changing the internal implementation. This makes a simple question surprisingly hard: after an And-Inverter Graph (AIG) has been optimized, when can a reviewer trust that an internal source expression, boundary, or cut has been recovered in the optimized netlist? This paper presents a research artifact that treats internal correspondence as a hierarchy of evidence rather than a single yes/no result. The artifact separates structural survival, semantic region recognition, exact locality certificates, graph-active rewriting, and global equivalence checking; it also keeps controlled, blind, oracle, generated, historical, and diagnostic denominators separate. The main empirical result is a recoverability frontier: controlled source-side counterparts and cross-netlist transplants reach global-CEC-backed graph-active recovery on their positive generated cases (10/10 and 12/12), while blind bounded semantic CEGIS recovers 3/24 regions, necessity-first generated targets expose 31/48 compact exact input interfaces and now emit 31/48 valid rewrite artifacts, of which 22/48 are graph-active CEC-backed new boundaries after bounded fanout-frontier expansion. Corrected historical diagnostic rows still support 0/56 graph-active recoveries. Within the controlled cross-netlist experiment, relational interfaces also expand accepted graph-active recovery from 9/17 direct-adapter rows to 12/17 relational-interface-enabled rows, showing that the boundary language itself can change recoverability. The negative results are not incidental: they are replayable blocker certificates for missing provenance, target irrelevance, non-compact locality, and absent rewrite artifacts. A new evidence-advancement layer records exactly which future-work rows move up in evidence level today: 0/56 source-blind rows reach graph-active recovery, 22/48 necessity-first rows become graph-active CEC-backed new boundaries under the bounded truth-table rewrite language, 4/12 operator/mode CEGIS groups are complete for their attempted rows, 3/3 tiny CC0 RTL designs are pinned but not lowered locally because Yosys is unavailable, 0/10 contextual ODC anchors reach graph-active placement, and 57/57 locality certificates have JSON proof objects. The repository therefore contributes an auditable framework for making internal-correspondence claims after logic optimization without mixing evidence levels or denominators.

1. Introduction

Modern logic synthesis is effective because it is not loyal to the internal structure of the source design. A synthesis flow can balance paths, rewrite AIG subgraphs, resubstitute nodes, exploit don't-care freedom, and then prove that the transformed circuit is equivalent to the original. For ordinary synthesis quality this is exactly the point. For debug, explanation, proof-carrying transformations, incremental ECO-style reasoning, source mapping, and optimization accountability, the same behavior is a problem. Whole-design equivalence says that the primary outputs agree; it does not say where a particular source-level computation survived, whether a recovered optimized node is useful at a local boundary, or whether replacing it would preserve the design.

This artifact studies that gap for AIG optimization experiments. The initial temptation was to look for a large internal recovery count: compare source and optimized nodes, rank candidates by structural and simulation similarity, and ask how many can be validated. The committed evidence shows that this framing is too coarse. Exact anchors are useful sanity checks, but non-exact structural candidates collapse under ABC equivalence checking. Semantic expression proofs can recognize isolated functionality, but isolated recognition does not imply a legal graph rewrite. Exact minimum locality certificates can prove that a compact interface exists, but an exact interface is still not a recovered boundary unless a graph-active artifact is emitted and the full circuit remains equivalent.

The central thesis is therefore:

Internal correspondence after AIG optimization has a recoverability frontier: structural similarity, semantic recognition, exact locality, graph activity, and global equivalence form distinct evidence levels, and serious claims must report where each denominator stops.

Figure 1 summarizes the motivating distinction. A synthesized design may pass whole-design CEC even when internal nodes no longer have a direct counterpart. The artifact asks which additional evidence promotes a candidate from a similar-looking internal signal to a recovered, graph-active correspondence.

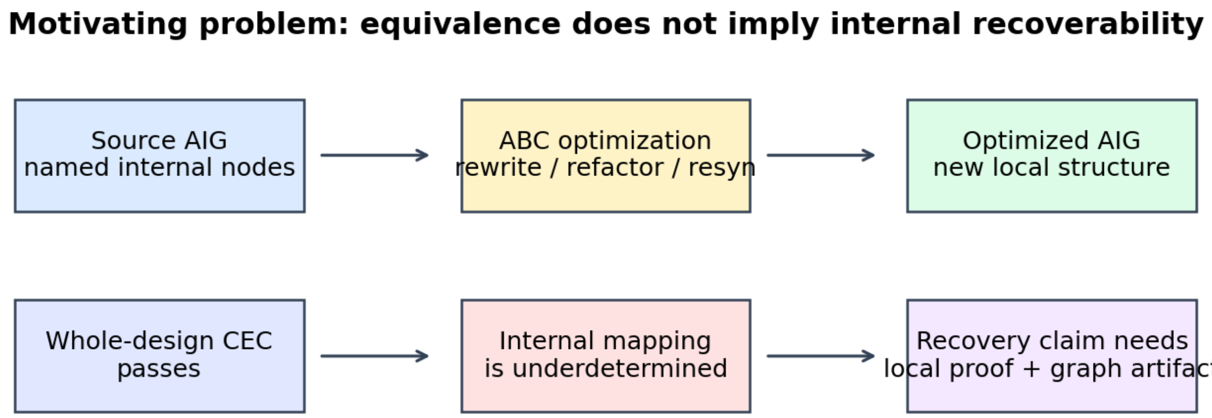

Figure 1: Motivating problem: whole-design equivalence does not imply internal recoverability

The contribution is not a new synthesis optimizer. It is a disciplined experimental framework for internal-correspondence recovery. Its novelty is in the way it combines proof obligations, failure accounting, and artifact validation:

1. **A recoverability evidence hierarchy.** The artifact defines and enforces separate levels for structural survival, semantic region recovery, exact locality certificates, graph-active rewrite artifacts, and global CEC.
2. **Constructive controlled recovery paths.** Source-side counterparts and cross-netlist transplants are accepted only after local proof, quotient or adapter construction, graph activity, and whole-design CEC. The committed controlled positives reach 10/10 and 12/12 accepted recoveries, respectively.
3. **Interface abstraction as a recovery parameter.** Cross-netlist transplantation shows that recoverability depends on the allowed boundary language, not only on node semantics: direct adapters recover 9/17 controlled new boundaries, while relational-interface-enabled transplantation recovers 12/17 under the same controlled denominator.
4. **Blind and oracle separation.** Blind bounded CEGIS, Z3-backed diagnostic CEGIS, oracle-bus rows, and oracle-window decomposition rows are reported as different evidence classes. Blind recovery is 3/24 in the primary parametric-CEGIS table, while the larger Z3 experiment is reported as a separate solver-scalability diagnostic.
5. **Compact-locality-to-rewrite promotion.** The necessity-first path now converts compact exact interfaces into concrete BLIF rewrite artifacts under a bounded truth-table rewrite language. It emits 31 valid artifacts; a bounded fanout-frontier expansion stage promotes four additional compact rows, yielding 22 graph-active CEC-backed new boundaries. The remaining nine emitted artifacts remain valid but are classified as identical-driver rewrites rather than constructive boundary recovery.
6. **Negative results as certificates.** Historical and generated failures are not treated as uninformative misses. They are classified by replayable blockers: missing optimized artifacts, target irrelevance after reconstruction, non-compact exact interfaces, no globally anchored cut, and non-active rewrite artifacts.
7. **A reviewer-oriented artifact contract.** Stable Make targets build paper tables and figures from committed CSVs, validate row-count freshness, run portable no-ABC checks, run ABC-backed checks when available, and compile this paper into a PDF.
8. **Evidence advancement without count inflation.** The repository includes a checked next-step layer that pins a redistributable RTL seed corpus, emits machine-checkable locality proof objects, and records which source-blind, compact-interface, CEGIS, and ODC rows still lack the obligations needed for graph-active claims.

2. Background and Related Work

The artifact is built around AIGs, ABC, SAT/SMT equivalence checking, and functional decomposition. AIGs are a compact Boolean circuit representation using two-input AND nodes and complemented edges; the AIGER format documents a portable file representation for such graphs (Biere 2007). ABC is the reference implementation used in this artifact for AIG optimization and combinational equivalence checking; the ABC paper describes it as a synthesis and formal-verification system based on scalable AIG transformations (Brayton and Mishchenko 2010).

The optimization flows studied here are directly related to DAG-aware AIG rewriting, balancing, refactoring, resubstitution, and don't-care based resynthesis. DAG-aware AIG rewriting reduces area by considering alternative subgraphs in a shared DAG, rather than optimizing each local expression in isolation (Mishchenko et al. 2006). FRAIGs integrate simulation and SAT reasoning to merge functionally equivalent AIG nodes and provide a common representation for synthesis and verification (Mishchenko et al. 2005). Don't-care based resynthesis further changes internal implementation by using local flexibilities and SAT-based care computation to justify replacements (Mishchenko et al. 2011). These methods are designed to improve circuits, not to preserve debug-friendly internal identity.

The artifact's proof obligations are aligned with equivalence-checking practice but applied to internal correspondence claims. Whole-design CEC asks whether the primary outputs of two designs are equivalent. Internal correspondence recovery asks for more: a candidate region, interface, or replacement must be locally well-defined and globally harmless. ABC supplies the CEC backend for netlist-level acceptance, while Z3 supplies bit-vector SMT proofs for the semantic CEGIS path (Moura and Bjørner 2008).

Counterexample-guided inductive synthesis is the closest methodological relative for the blind semantic path. The classic sketching work uses a candidate-generation and verification loop over finite programs (Solar-Lezama et al. 2006). Here, the same pattern is adapted to bounded hardware-region templates: propose a candidate expression from examples, prove or refute it against the region truth table or SMT formula, and add a concrete counterexample when it fails.

Functional decomposition provides the conceptual basis for quotient-style refactoring. Ashenhurst decomposition and Curtis's later formulation study whether a Boolean function can be expressed through intermediate subfunctions over smaller variable sets (Ashenhurst 1957; Curtis 1962). The artifact uses this idea operationally: a recovered semantic divisor is not useful until the remaining source or target window admits an exact quotient and the resulting graph edit survives global CEC.

Benchmarks include generated BLIF families and ISCAS-85-style netlists. The ISCAS-85 circuits remain a common combinational benchmark reference (Brglez and Fujiwara 1985), but this artifact is careful not to treat every legacy or historical result row as a valid denominator. A row is a valid denominator only when its source artifact, optimized artifact, target, interface, proof scope, and checker contract are reconstructible.

3. Problem Formulation

Let $G_s = (V_s, E_s)$ be the source AIG and $G_o = (V_o, E_o)$ the optimized AIG produced by a synthesis flow. Both graphs have primary inputs I and primary outputs O. A conventional equivalence check proves

$$\forall i \in \{0,1\}^{|I|}.\ G_s(i)|_O = G_o(i)|_O.$$

This relation is necessary for any graph rewrite accepted by the artifact, but it is not sufficient for internal correspondence.

An **internal correspondence candidate** is a tuple

$$c = (S, T, B_i, B_o, \phi, e)$$

where S is a source-side expression or region, T is an optimized-side target or region, B_i and B_o are input and output boundary interfaces, ϕ is an optional adapter or quotient, and e records the evidence level. Candidates may be source-blind, oracle-assisted, or controlled:

- **Source-blind** candidates may use optimized graph structure, simulation, CEGIS examples, and public benchmark metadata, but not the hidden ground truth expression at inference time.
- **Oracle** candidates may use selected ground truth to diagnose whether a failure is due to interface discovery, factorization, or implementation.

- **Controlled** candidates are generated with known source counterparts and are used to test whether the proof and rewrite machinery can accept valid positives and reject invalid controls.
- **Historical diagnostic** candidates are previous rows whose provenance is audited after the fact; they are not automatically eligible recovery attempts.

The artifact defines six evidence levels, shown in Figure 2. The cross-netlist ablation motivates the extra interface level: a recovered semantic function is not necessarily recoverable through a particular boundary language.

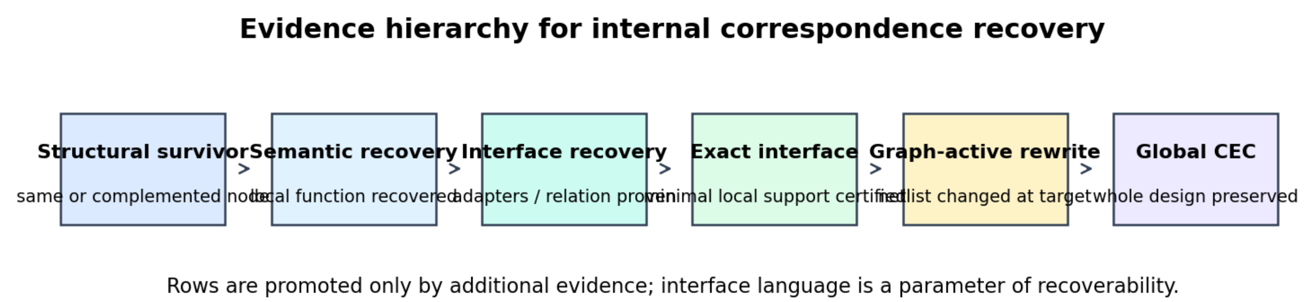

Figure 2: Evidence hierarchy for internal correspondence recovery

Structural survival. A node is structurally recovered when a same-polarity or complemented candidate is validated against a source node under the chosen scope. This is a useful sanity layer but does not establish that a source-level boundary can be rewritten.

Semantic region recovery. A region is semantically recovered when a candidate expression is formally equivalent to the region function under the declared interface. This level supports recognition claims.

Interface recoverability. A candidate is interface-recoverable when the semantic region can be connected to the surrounding source and optimized context using an admissible adapter language. This level is stricter than semantic recognition and weaker than graph-active rewriting.

Exact locality. A candidate has an exact local interface when the artifact proves a lower bound and matching upper bound for the required boundary within the declared universe. This level supports locality claims, not graph-rewrite claims.

Graph-active recovery. A graph rewrite is graph-active when it changes the netlist at the target rather than reintroducing an identity or dead artifact. The artifact does not count local proofs or materialized diagnostic wires as graph-active recoveries unless a concrete rewrite artifact is emitted.

Global CEC-backed recovery. A graph-active rewrite is accepted only when ABC proves the full source/optimized or cross-netlist design equivalent after the edit.

For clarity, the paper uses $R_L(t)$ for the recoverability of target t under an admissible interface language L. Here L_{direct} permits direct exact adapters over the declared boundary variables, while L_{rel} additionally permits the controlled relational-interface path recorded by the latent interface tables. In the 17-case controlled cross-netlist ablation, $R_{L_{direct}}$ accepts 9 targets and $R_{L_{rel}}$ accepts 12. This is a language-parameterized frontier, not a universal claim about all possible interface formalisms.

The research questions follow from the hierarchy:

RQ1. How quickly do structural internal correspondences erode under common AIG optimization flows?

RQ2. How far can blind bounded semantic CEGIS recover internal region semantics without oracle access?

RQ3. When a source-side semantic counterpart is known or constructible, can the artifact turn it into graph-active CEC-backed recovery?

RQ4. Can cross-netlist adapters move controlled correspondences between optimized designs, and does the allowed interface language change the recoverable set?

RQ5. What do the null results say about locality, provenance, and target necessity in generated and historical rows?

4. Framework

The implementation is organized as three methodology lanes and one shared acceptance discipline (Figure 3). The blind lane estimates what can be inferred without privileged source expressions. The controlled lane checks whether valid semantic correspondences can be turned into graph-active edits. The audit lane explains why rows from earlier experiments should or should not count as eligible denominators.

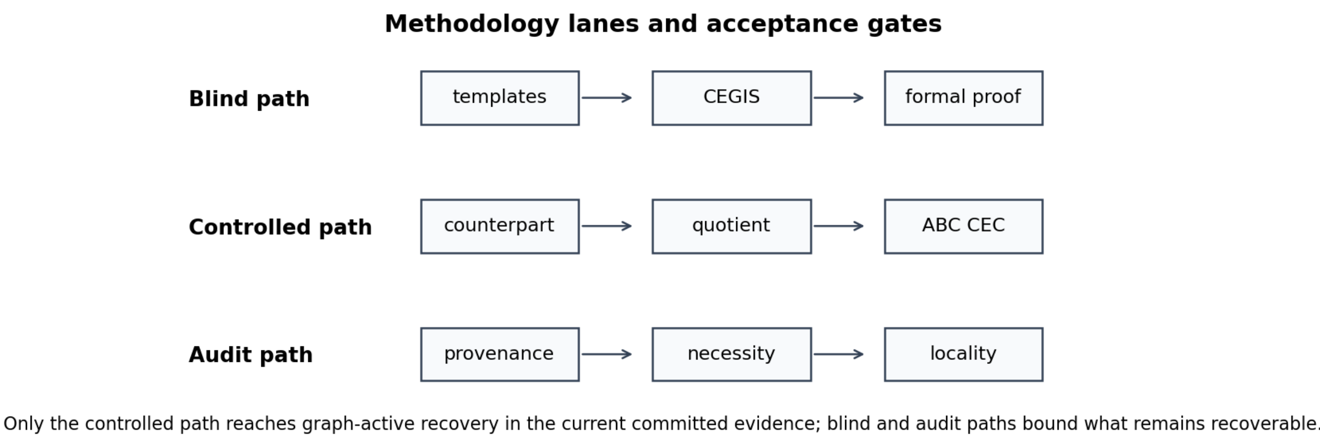

Figure 3: Methodology lanes and acceptance gates

The shared acceptance discipline is deliberately strict:

1. Build or load the source and optimized BLIF/AIG artifacts.
2. Identify a candidate target and boundary universe.
3. Prove the local semantic relation, or record a concrete counterexample.
4. If the claim is a rewrite claim, emit a concrete graph artifact and show that the artifact changes the target.
5. Run ABC CEC over the full relevant design pair.
6. Record the result with `schema_version`, result family, denominator class, evidence level, source table, and checker.

This discipline prevents three common overclaims. First, a formally verified semantic expression is not counted as a boundary restoration unless it has a legal interface and graph placement. Second, an exact minimum interface is not counted as a rewrite. Third, a historical row is not counted as a failed or successful attempt under the final algorithm unless its artifacts and target can be reconstructed under the current contract.

4.1 Blind Semantic CEGIS

The blind CEGIS path treats each region as an unknown Boolean or bit-vector function over an inferred interface. It enumerates templates such as constants, linear forms, affine variants, arithmetic templates, or simple bit-manipulation operators. For each candidate, the verifier either proves equivalence or returns an assignment that distinguishes the candidate from the region. The assignment is appended to the example set and the loop continues.

The important artifact property is replayability. Each counterexample row stores the candidate, examples-before/examples-after counts, solver status, and assignment. The checker verifies that SAT refinement rows increase the example set. Thus a blind failure is not merely “no result”; it is a bounded search trace showing how candidates were eliminated.

4.2 Source-Side Counterpart Refactoring

The source-side counterpart path assumes a controlled setting where a candidate source expression can be materialized or constructed. The artifact then asks whether this source-side signal can be used as a divisor for the target function. A positive row must prove:

- the counterpart is locally equivalent to the intended semantic function;
- the target window is decomposable through that counterpart;
- the quotient depends non-vacuously on the counterpart;
- the emitted graph edit is non-identity and graph-active;
- source-side and cross-netlist ABC CEC both pass.

This is the cleanest positive path in the current artifact because it connects semantic proof, quotient synthesis, graph activity, and global equivalence.

4.3 Cross-Netlist Cut Transplantation

Cross-netlist transplantation generalizes the controlled source-side path to move a cut across related optimized netlists. It constructs input adapters, optional relational interfaces, output adapters, and then performs graph-active replacement under global CEC. The controlled experiment includes negative controls, so acceptance is not just the result of overly permissive checking.

The direct adapter language treats the connection as a deterministic exact truth-table map from the declared source-side boundary variables and residuals to the optimized cut inputs, and from the optimized cut outputs plus residuals back to the source outputs. Formally, the implementation accepts an adapter when all assignments that agree on the adapter interface agree on the adapter output; otherwise it records a two-copy counterexample. This is a functional adapter model over a fixed boundary.

The relational-interface path allows the transplantation procedure to introduce and prove a small latent relation over the boundary before graph replacement. In the committed controlled experiment this is represented by a one-bit latent interface `k0` with `proof_status=proved` and `formal_status=equivalent` in the relational-interface tables. The extra expressive power is not a different CEC standard: accepted relational rows still require exact input/output adapters, local equivalence, graph activity, and both ABC CEC scopes. It changes the boundary language available before those obligations are checked.

The ablation structure matters: direct adapters produce fewer accepted boundaries than relational-interface-enabled transplantation, while GF(2) linear special cases are diagnostic and do not solve the general problem.

4.4 Necessity and Locality Auditing

The necessity-first audit begins with optimized targets that have sufficient provenance. It filters targets by non-constancy, forced observability, and reachable necessity. The locality checker then proves whether a compact exact input interface exists under the declared universe. Finally, graph rewrite accounting records whether a validated rewrite artifact was emitted.

The audit path is where the artifact is most conservative. A target with a compact exact interface is still not counted as recovered unless the rewrite artifact exists. Historical rows with missing optimized artifacts or target irrelevance are removed from eligible denominators and reported as diagnostics.

5. Experimental Methodology

The repository contains several result families rather than a single monolithic experiment. This section defines the common methodology used by the paper.

Toolchain. ABC is pinned through the Makefile variable `ABC_REV`; the reviewer-safe checks can run without ABC, while formal targets build or use the pinned ABC binary. Z3-backed paths use the Python `z3-solver` dependency, and the Z3 cross-check table records Z3 version 4.16.0 in the committed rows. Pandoc and LaTeX compile this paper into the repository's paper-output directory.

Benchmarks. The artifact includes generated BLIF benchmarks, generated semantic-recovery benchmark families, and ISCAS-85-style external netlist diagnostics. It also now includes a tiny redistributable CC0 RTL seed corpus with source-location metadata. This is not an RTL-recovery denominator: local validation records Yosys as unavailable, so no successful RTL lowering or RTL-to-netlist correspondence claim is made.

Denominator classes. The paper uses the following denominator labels:

Denominator class	Meaning	Counted together?
`controlled_generated_blif`	generated positives/controls with known construction intent	only with controlled rows
`blind_generated_blif`	source-blind semantic inference rows	only with blind rows
`generated_research_benchmark`	provenance-complete generated targets	only with necessity-first rows
`historical_diagnostic`	audited previous rows with diagnostic evidence	never as final rewrite attempts
`historical_ineligible_diagnostic`	previous rows excluded by provenance or target necessity	never as recovery attempts
`oracle_*`	rows using privileged information for diagnosis	never merged with blind rows

Evidence and freshness. Result tables carry schema versions and are checked by family-specific scripts. `make artifact-check` rebuilds the artifact manifest, validates committed result freshness, checks all major paper-facing claims, and runs the research-wow generator. `make reproduce-paper-tables` regenerates the tables and figure used here and validates that the claims still match the committed CSVs.

6. Results

6.1 Structural Similarity Is Not Recovery

The first result is a cautionary baseline. The repository has a structural and SAT-refinement layer that checks direct exact anchors separately from non-exact candidate recovery. The committed SAT validation-layer summary contains 3052/3052 exact-anchor sanity validations, but 0/425 rank-1 non-exact recoveries and 0/1993 top-k non-exact recoveries under ABC validation. The complemented follow-up also finds 0/425 rank-1 and 0/1993 top-k complemented recoveries.

This does not mean that all internal information disappears. The broader summary-metrics diagnostic table shows nontrivial exact-match rates for some low- and medium-effort flows. It means that similarity metrics alone are not enough to support internal-correspondence claims after aggressive optimization. They are ranking features, not proof artifacts.

6.2 Blind Bounded Semantic Recovery Is Real but Narrow

The primary blind semantic result is intentionally small. In the blind semantic recovery summary, the blind parametric CEGIS mode attempts 24 regions, formally verifies 3, records 47 iterations, adds 23 counterexamples, and reports zero timeouts. This is a 12.5% formal recovery rate under the primary blind denominator.

The artifact also includes a larger Z3-backed diagnostic CEGIS experiment. In that comparison table, both blind and oracle-bus modes attempt 16 unique cases and recover 10; both recover all four attempted 12/16-bit wide cases. This result is useful for solver scalability and template expressiveness, but it is not merged with the primary 3/24 blind frontier because it is a different experiment with different rows and proof backend. The Z3 exhaustive cross-check provides an independent sanity layer: 192/192 supported small candidate checks agree between exhaustive validation and Z3.

The scientific reading is that blind CEGIS can produce formally checked semantic recognitions and replayable failures, but bounded templates do not yet recover most regions under the primary blind setting.

6.3 Isolated Semantic Proofs Do Not Imply Graph Placement

The graph-placement phases explain why recognition alone is insufficient. The semantic-grafting summary reports 46 proven semantic expressions and 276 bounded placement attempts, with 0 accepted graph-active semantic grafts. The failure taxonomy repeats the same core message across placement attempts: no exact observable output context frontier, no mapped cut leaves, no legal fanout edge, and whole-design expansion risk.

The controlled region-replacement phases show that the machinery is not incapable of accepting positives. The semantic-region replacement summary reports 6 SMT-verified free-cut semantic modules, 7 replacement attempts, and 5 accepted graph-active replacements. The joint region/interface summary reports 10 controlled cases, 9 verified modules, 8 graph-active replacements, and 8 restored controlled boundaries. The semantic functional-refactoring summary reports 13 controlled experiments, 12 decomposable candidates, 12 exact quotients, 10 non-identity accepted decompositions, and 10 graph-active global-CEC replacements. The same family reports 58 real development/held-out attempts and 0 real restored boundaries.

These rows locate the frontier: controlled semantic construction can work, but real isolated anchors do not automatically provide closed implementation regions.

6.4 Controlled Source-Side Counterparts Reach Graph-Active Recovery

The strongest positive source-side result is in the active source-counterpart controlled table and the derived recoverability-frontier table. The controlled positive denominator is 10 and all 10 rows are accepted graph-active counterpart rewrites with source and cross ABC CEC. The final supported claims table keeps this separate from the 56 real/historical rows, which remain at 0 graph-active recoveries and 0 new boundaries.

The ablations explain why this is not simply a name-matching artifact. `old_target_selection` and `proof_easiness_ranking` each attempt 20 rows and recover 0 new boundaries. The boundary-utility-aware ranking ablation is the one that reaches 10 new boundaries. The GF(2) linear special case proves some local structure but remains a diagnostic baseline rather than the general refactoring algorithm.

6.5 Cross-Netlist Transplantation Benefits from Relational Interfaces

The cross-netlist controlled path reaches 12 accepted transplants in the paper-facing frontier. The underlying controlled table has 17 attempted rows including controls; 12 accepted rows are the controlled positive graph-active recoveries counted in the frontier. Historical development rows remain 0/56 for new recovered boundaries.

The ablation table is more informative than the headline alone. Direct adapters produce 9/17 new boundaries, while relational-interface-enabled transplantation produces 12/17. The relational row also increases graph-valid replacements from 12/17 to 15/17 and global CEC passes from 10/17 to 13/17. The GF(2) linear relational baseline attempts 34 rows and recovers 0 new boundaries, showing that the effect is not captured by a narrow affine special case.

Cross-netlist interface model	Relational rows proved	Graph-valid	Global CEC	New boundaries
Direct adapters only	0	12 / 17	10 / 17	9 / 17
Relational-interface enabled	3	15 / 17	13 / 17	12 / 17

The three additional accepted rows are not anonymous increments. They are the controlled `xor_basis_adapter`, `nonlinear_boolean_adapter`, and `bilinear_transplant` cases. The committed relational-interface tables record all three with `latent_width=1`, `latent_interface=["k0"]`, `proof_status=proved`, and a matching latent-interface proof with `formal_status=equivalent`. The controlled result rows then show exact input and output adapters, graph-active cloned regions, local exhaustive equivalence, source CEC equivalence, cross CEC equivalence, and `new_recovered_boundary=true`.

The implementation clarifies what these rows mean. The `xor_basis_adapter` source cut computes the optimized basis as `(a xor b, b)` rather than a plain wire-aligned cut. The `nonlinear_boolean_adapter` compresses four source boundary bits into two optimized cut inputs `(a and b, c or d)`. The `bilinear_transplant` row uses the relational path for a bilinear target interface. In all three cases the accepted evidence is carried by the relational-interface path and then discharged by the same local-proof and CEC obligations as the direct rows.

This supports a stronger interpretation than "three more cases succeeded." In this controlled 17-case experiment, the admissible interface language changes the recoverability frontier. Some correspondences are blocked under a restricted boundary representation even though an exact cross-netlist relationship is available in the richer relational representation used by the artifact. The current tables do not prove that every conceivable direct encoding of these three rows is impossible; they do prove that the implemented direct-only frontier is strictly smaller than the implemented relational frontier under the same denominator.

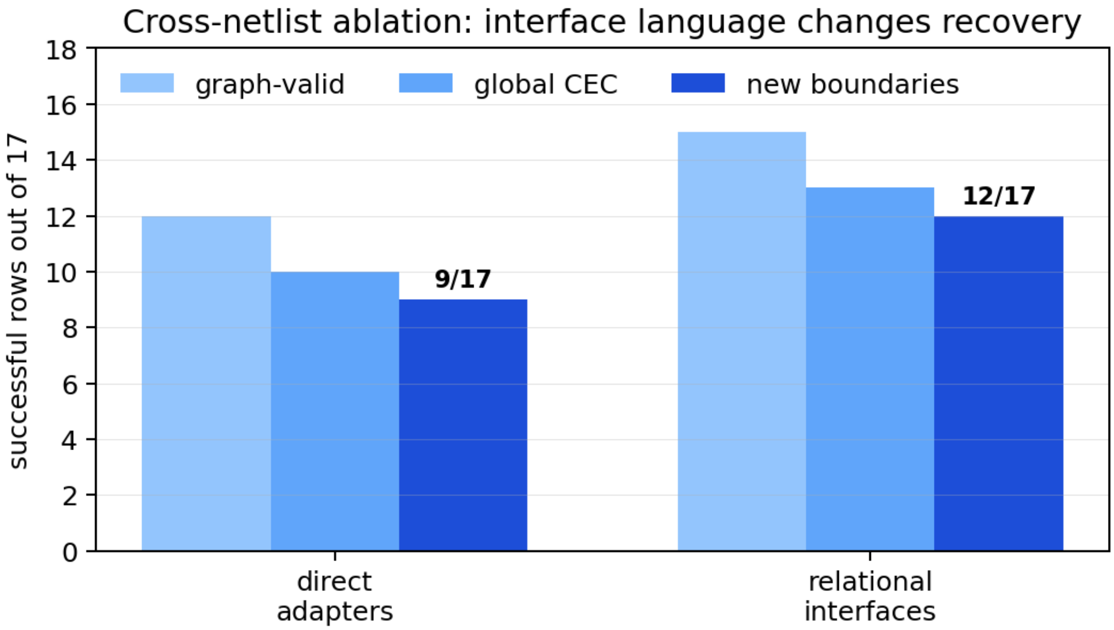

Figure 4: Cross-netlist ablation: interface language changes recovery

6.6 Necessity-First Auditing Promotes Compact Locality to Rewrites

The necessity-first target discovery phase is the artifact's clearest example of claim discipline. It identifies 48 provenance-complete generated targets that pass the necessity-first filter. The locality checker proves compact exact input interfaces for 31/48. If the paper stopped there, it would overclaim: those 31 rows are exact-locality certificates, not graph-active recoveries.

The bounded rewrite-synthesis stage first reconstructs each compact target as a single-output truth table over its certified interface, replaces the optimized target driver in BLIF, preserves existing fanouts, and then validates graph shape and both CEC scopes. It emits 31/48 valid rewrite artifacts. In the single-output language, 18/48 are graph-active and CEC-backed new boundaries. The artifact then applies a bounded fanout-frontier expansion to the remaining CEC-equivalent but non-active rows: it replaces the target and one immediate fanout consumer when the expanded frontier stays within radius 1, at most two replacement outputs, and at most six truth-table inputs. This promotes 4 additional rows, all adder carry-propagation frontier cases, giving 22/48 graph-active CEC-backed new boundaries. The remaining 9 emitted artifacts pass CEC but are classified as identical-driver rewrites; the attempted expansion would make the target no longer reach an output under the configured bound.

Thus the correct statement is:

> Necessity-first generated targets expose many compact exact interfaces (31/48), all compact rows now emit valid rewrite artifacts (31/48), and the bounded rewrite language promotes 22/48 to graph-active CEC-backed new boundaries, including 4 rows that require fanout-frontier expansion beyond the single-output target driver.

This distinction is central to the paper. It turns a potentially confusing null result into a precise research finding: local semantic sufficiency can be made constructive for a substantial subset, but artifact emission and graph-active boundary recovery remain distinct evidence levels.

6.7 Historical Rows Are Diagnostic, Not a 56-Row Recovery Denominator

The provenance eligibility audit corrects an older stale contract. The 56 historical rows are not eligible graph-rewrite attempts under the current artifact contract. The provenance reconstruction table contains 36 rows with missing optimized artifacts and 20 rows whose provenance can be reconstructed exactly but whose historical target is not necessary for the selected interface. The corrected eligible historical graph-rewrite denominator is therefore 0.

This correction is important because it prevents two opposite errors. The artifact should not claim successful historical recovery from rows that lacked reconstructible artifacts. It also should not count those rows as 56 failures of the final algorithm. They are evidence for provenance and target-necessity barriers.

6.8 The Recoverability Frontier

The main paper-facing table is generated by `make research-wow` from committed CSV inputs and checked by `make check-research-wow`.

Result family	Denominator class	Success / denominator	Evidence level
Controlled active source counterparts	controlled generated BLIF	10 / 10	formal exhaustive + ABC CEC
Controlled cross-netlist transplants	controlled generated BLIF	12 / 12	formal exhaustive + ABC CEC
Blind parametric CEGIS	blind generated BLIF	3 / 24	formal exhaustive
Necessity-first compact interfaces	generated research benchmark	31 / 48	exact minimum certificate
Necessity-first graph rewrites	generated research benchmark	22 / 48	truth-table rewrite + fanout-frontier expansion + ABC CEC
Formal locality previous failures	historical diagnostic	26 / 56	exact minimum certificate diagnostic
Historical cross-netlist recovery	historical ineligible diagnostic	0 / 56	corrected denominator audit

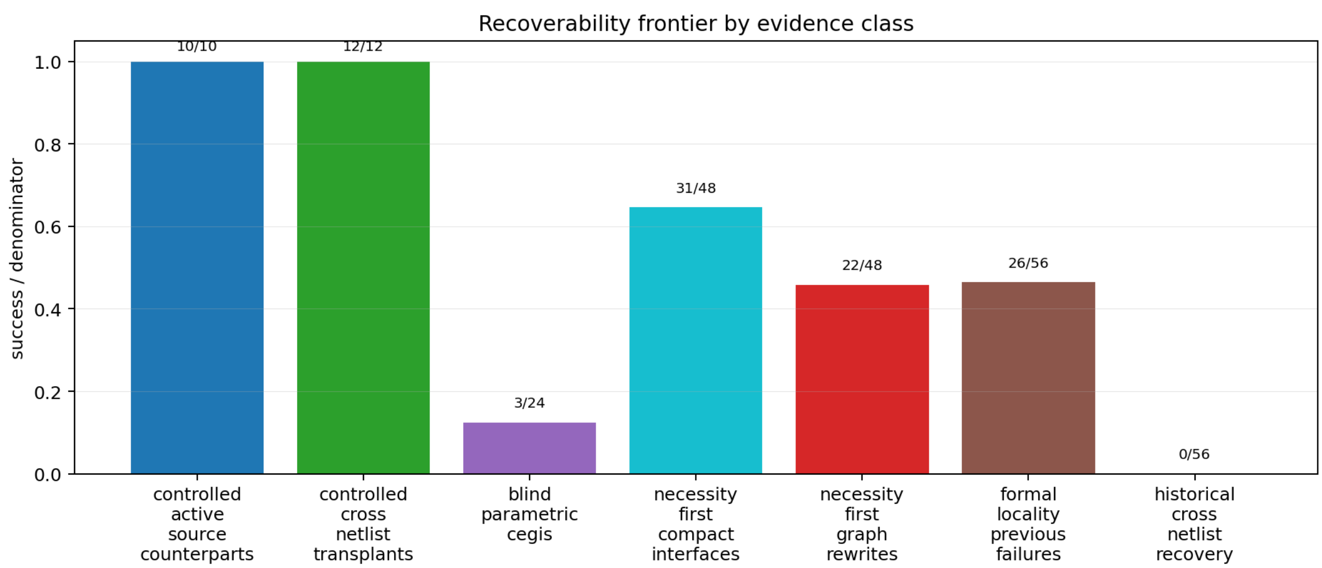

Figure 5: Recoverability frontier by evidence class

The table is deliberately heterogeneous because the artifact's main claim is about heterogeneity. The figure does not invite aggregation; it displays which kind of evidence each count belongs to. The cross-netlist ablation adds a second dimension: even within one controlled denominator, the frontier is parameterized by the admissible interface language L. The result $|R_{L_{direct}}| = 9$ and $|R_{L_{rel}}| = 12$ for the 17 controlled rows means recoverability is partly a property of the chosen boundary abstraction, not only of the underlying Boolean function.

6.9 Failure Taxonomy

The failure taxonomy turns null results into research data (Figure 5). The largest top-level classes include 48 generated targets without validated graph rewrite artifacts, 36 historical rows with missing optimized artifacts, 36 real active-source or cross-netlist rows without globally anchored cuts, 21 blind CEGIS bounded-exhaustion failures, 20 target-irrelevant reconstructed historical diagnostics, and 17 generated targets with non-compact exact input interfaces.

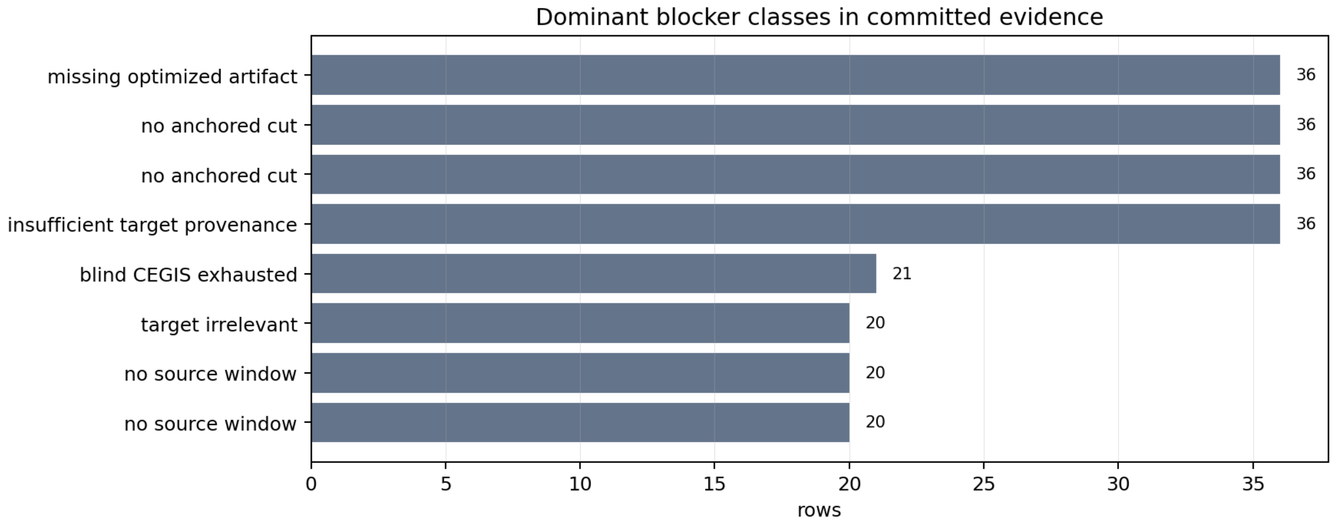

Figure 6: Dominant blocker classes

These categories provide an actionable map for future work. The artifact does not merely say that recovery failed. It says whether the blocker was evidence provenance, target relevance, blind template expressiveness, local interface size, placement legality, or missing graph artifact emission.

7. Case Studies

Figure 6 shows the two representative traces generated by `make demo-wow`.

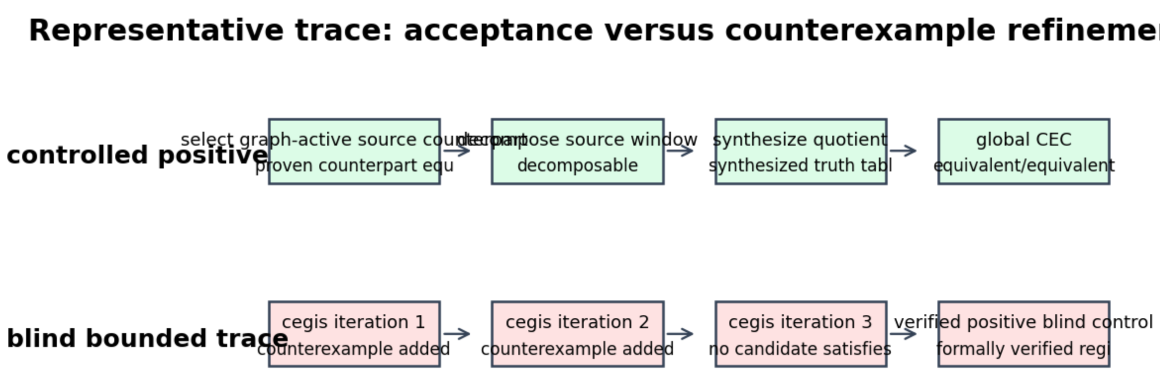

Figure 7: Representative trace: acceptance versus counterexample refinement

The controlled half follows an accepted source-counterpart case. The row proves a graph-active source counterpart, decomposes the source window, synthesizes the quotient, and checks global source/cross equivalence with ABC. The key point is that the acceptance is not a single proof obligation; it is a chain of local and global obligations.

The blind half follows a bounded CEGIS trace for an arithmetic add-add region. The first candidate is refuted by a concrete counterexample, the example set is expanded, a second candidate is refuted, and the third row records bounded exhaustion for that region. A separate positive blind control in the same demo has a formal proof. This pairing is useful for reviewers because it shows both sides of the blind claim: verified positives and replayable negative refinement.

8. Discussion

The artifact suggests three lessons for research on internal correspondence.

First, **correspondence must be typed by use**. If the downstream use is a debug hint, a structural or approximate candidate may be useful. If the use is proof-carrying graph rewrite, local similarity is irrelevant without graph activity and global CEC. The artifact's evidence hierarchy makes the intended use explicit.

Second, **negative results can be stronger than weak positives**. A failed blind CEGIS row with a replayable counterexample trace is more scientifically useful than an unverified high-similarity match. A historical row audited as missing an optimized artifact protects the paper from claiming either success or failure under the wrong denominator.

Third, **oracle diagnostics are valuable only when labeled**. Oracle divisor, oracle support, and oracle window rows can identify whether the remaining blocker is decomposition, locality, or graph placement. They become misleading only when merged with blind recovery rates. The artifact keeps them separate.

Fourth, **some barriers are abstraction-induced**. A failed attempt under a restricted interface model should not automatically be read as evidence that no compact or useful correspondence exists. In the controlled cross-netlist ablation, enabling the relational-interface language expands graph-active recovery from 9/17 to 12/17. Those three rows indicate that a boundary can be too restrictive even when the internal function is recoverable and a richer cross-netlist relation can be proven. This is different from the formal locality-barrier rows where exact minimum-interface certificates establish that no compact interface exists under the configured universe and bound. The first case asks for a richer language; the second asks for a larger or different locality universe.

The current controlled positives are also not the endpoint. They show that the acceptance machinery can recognize valid graph-active transformations. The open research challenge is to close the gap between controlled construction and source-blind real recovery.

9. Threats to Validity

Benchmark scope. The artifact uses generated BLIF and netlist benchmarks with some ISCAS-85-style diagnostics. It does not yet support broad external RTL claims. A paper using this artifact should not imply that the results generalize to industrial RTL without adding a pinned redistributable corpus and lowering flow.

Flow scope. The optimization passes are representative AIG flows, not an exhaustive synthesis-space study. The recoverability frontier could shift under different ABC revisions, custom scripts, technology mapping, or sequential optimizations.

Bounded search. Several negative results are bounded. The artifact records the bound and blocker class; it does not claim mathematical impossibility outside the declared universe unless an exact lower-bound certificate says so.

Controlled denominators. Controlled generated positives are designed to exercise the proof-carrying machinery. They should not be reported as blind real-benchmark success rates. Their purpose is to show that the acceptance contract is meaningful and non-vacuous.

Historical data. Historical rows are valuable for auditing provenance and target relevance, but the current paper treats them as diagnostics after the audit corrected their denominator. They are not final-algorithm attempts.

10. Reproducibility

The artifact exposes stable reviewer targets:

```
make smoke
make portable-no-abc
make evidence-advancement
make check-evidence-advancement
```

```
make artifact-check
make reproduce-paper-tables
make paper-pdf
```

When ABC is available or can be built:

```
make formal-abc
```

The PDF target regenerates the research-wow tables and figures before invoking Pandoc. The resulting PDF is written to the paper-output directory. The artifact manifest records result families, commands, row counts, hashes, git state, Python version, ABC path/revision metadata where available, and schema versions.

11. Future Work

The next research step is not to inflate current counts. It is to move rows across evidence levels honestly. The repository therefore adds `results/evidence_advancement/`, generated by `make evidence-advancement` and validated by `make check-evidence-advancement`. The current checked advancement state is:

Direction	Current promoted rows	Honest interpretation
Source-blind source-side counterpart inference	0 / 56	20 rows have semantic-only counterpart evidence, but no row has a new graph-active recovery.
Graph-active rewrites from compact generated interfaces	22 / 48	31 compact exact interfaces emit valid rewrite artifacts; fanout-frontier expansion promotes 4 additional rows to graph-active CEC-backed new boundaries.
Bounded grammar completeness for selected CEGIS families	4 / 12	Only `sign_extend` and `zero_extend` are complete for their attempted blind and oracle-bus rows.
Pinned redistributable RTL corpus	3 / 3	Three CC0 Verilog modules and source metadata are committed; local Yosys lowering is `tool_missing`.
ODC-aware placement	0 / 10	Ten formal contextual ODC anchors exist, but none is counted as graph-active without global CEC.
Machine-checkable locality proof objects	57 / 57	JSON proof objects mirror the exact-minimum locality CSV certificates.

This layer makes the next paper-worthy work precise. The direct engineering targets are a source-blind counterpart synthesizer, broader fanout and multi-output rewrite languages for the remaining identical-driver rows, broader CEGIS grammars with completeness proofs for selected operator families, an installed and pinned Yosys lowering flow for the new RTL seed corpus, ODC-aware graph placement with explicit graph-activity and global CEC obligations, and richer proof objects that go beyond mirroring replayable CSV certificates.

12. Conclusion

Internal correspondence after logic synthesis is recoverable only under the right evidence contract. This artifact demonstrates that contract on AIG optimization experiments. Controlled source-side counterparts and cross-netlist transplants can reach graph-active CEC-backed recovery. Blind CEGIS can recover some bounded semantic regions and produce replayable counterexamples for failures. Necessity-first auditing can now prove compact exact interfaces, emit concrete rewrite artifacts, and promote a larger subset to graph-active CEC-backed recovery with fanout-frontier expansion while keeping identical non-active artifacts unpromoted. Historical rows can be corrected into provenance and target-necessity diagnostics instead of being misused as recovery denominators.

The result is a recoverability frontier: a map of what survived, what can be recognized, what can be localized, what can be rewritten, and what remains blocked. The cross-netlist ablation sharpens that map: richer relational interfaces expand the constructively recoverable set in the controlled experiment, while formal locality certificates remain necessary to distinguish representational limitations from genuine locality barriers. That language-parameterized frontier is the paper's scientific object and the repository's artifact contract.

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